higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 2

(12030152)

22 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: BASIC CONCEPTS AND ROAD-BUILDING TERMINOLOGY

1.1 Differentiate between the following earthwork activities:

1.1.1 Cut operations
1.1.2 Fill operations

(2 × 2) (4)

1.2 What do the following designated symbols, which are usually used for road construction materials, stand for:

1.2.1 'G'
1.2.2 'C'
1.2.3 'A'
1.2.4 'S'

(4 × 1) (4)

1.3 Name TWO different types of road furniture commonly found inside a road reserve. (2)

[10]

QUESTION 2: ROAD SURVEY

2.1 Explain the meaning of the following terms commonly used when setting out a new road.

2.1.1 Gradient
2.1.2 Centre line

(2 × 2) (4)

2.2 Name FIVE different types of hand tools and equipment that will be necessary when using the 3.4.5 (Pythagoras) method to set out a right angle on a construction site. (5)

2.3 State FOUR reasons why some inclined surfaces where a road is being constructed may fail. (4)

2.4 Explain the procedural steps that you will follow when using a string line and line level, to level out the surface of a road between two fixed profiles positioned on either side of the road. (5)

2.5 Answer the following in terms of a benchmark.

2.5.1 What is a benchmark?
2.5.2 Which 'mean' level is a benchmark measured from?

(2 × 1) (2)

[20]

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QUESTION 3: SOIL, GRAVELS AND ROAD-BUILDING MATERIALS

3.1 State the particle sizes and what the following road-building materials consist of:

3.1.1 Gravel
3.1.2 Sand
3.1.3 Clay
3.1.4 Silt

(4 × 2) (8)

3.2 State the mechanical properties of soil in terms of the following:

3.2.1 Permeability
3.2.2 Compaction

(2 × 2) (4)

3.3 State THREE reasons for spraying water onto soil during the compaction process of a road layer. (3)

3.4 Explain how you will conduct a hand test to evaluate and determine the value of the plastic limit (moisture content) in a sample of soil? (5)

[20]

QUESTION 4: INTRODUCTION TO ROAD DRAINAGE

4.1 List FIVE reasons for having an efficient road drainage system on the carriageway of a road. (5)

4.2 Slope or embankment surfaces are often protected by spreading a layer of topsoil over the surfaces of the slope or embankment.

Substantiate why a topsoil layer is sometimes used on some slope or embankment surfaces. (2)

4.3 Identify THREE aspects on how the following drainage systems differ from each other:

4.3.1 Surface drainage (3)
4.3.2 Subsurface drainage (3)

4.4 Name TWO types of materials that the following types of culverts can be made or composed of:

4.4.1 Flexible culvert (2)
4.4.2 Rigid culvert (2)

4.5 Name THREE types of underground drainage systems that are usually used. (3)

[20]

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QUESTION 5: EARTHWORKS

5.1 Earthworks involve operational activities that require reaching the formation level of the road, which forms the platform on which the pavement layers for the road will be constructed.

Name the TWO main operational activities that earthworks consist of. (2)

5.2 Discuss SIX different situations where excavation cave-ins are more likely to occur if the sides of the excavations are not protected against collapse. (6)

5.3 List FOUR responsibilities set out by the Machinery and Occupational Safety Act that the workers have to adhere to, to ensure their safety on site. (4)

5.4 List FIVE different types of plant and road construction machinery usually used for earthwork activities, such as moving and hauling on a road construction site. (5)

5.5 List THREE aspects that need to be considered when designing and planning an accessible route for large trucks to use when hauling and moving between stockpiles and work areas. (3)

5.6 Explain why cave-ins or collapse of trenches/excavations can be dangerous or even deadly. (2)

5.7 List THREE major stages of earthworks that will require specific techniques and equipment. (3)

[25]

QUESTION 6: LAYER WORKS

6.1 Name ONE type of road construction material that can be used as a base course layer immediately below the surface layer of a road. (1)

6.2 Differentiate between the following types of unpaved roads.

6.2.1 Earth road
6.2.2 Gravel road

(2 × 2) (4)

[5]

TOTAL: 100

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